

Acceptance of Simulated Body-Proximate Technologies in XR: A Transhumanist Perspective

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ABSTRACT

Current hardware developments exhibit a trend towards body-proximate technologies, moving beyond handheld devices to smart glasses, rings, and possibly bio-integrated sensors. As technology transitions from external tools into the user's immediate personal space and biological boundaries, the requirements for adoption shift; users should be able to trust the device, and feel a sense of ownership, perceiving it as an integral part of themselves, at the same time. By simulating body-proximate technology in Extended Reality (XR), we aim to understand how the degree of invasiveness influences technology acceptance and the psychological construct of embodiment. We propose a study design featuring three distinct scenarios representing an incremental scale of invasiveness, ranging from external wearables to structural body modifications. In this paper, we present the results of an explorative pre-study ($N = 37$) that evaluates user perceptions and the validity of these chosen scenarios. We discuss these preliminary findings and outline the methodological framework for a planned main study designed to investigate the interplay between invasiveness, embodiment, and user acceptance.

Index Terms: Transhumanism, Body-Proximate, XR, Extended Reality

1 INTRODUCTION

Current research trends exhibit a strong drive to improve the human body through technology. While the common approach has involved presenting information via Virtual Reality (VR) headsets or Head-Mounted Displays (HMDs), the trajectory of hardware development is moving closer to the biological body itself. Devices such as smartwatches are already established, yet newer form factors like bracelets and fitness rings continue this trend, utilizing multimodal approaches to record and evaluate biological data with increasing ease [9]. While the development of novel hardware requires significant manpower, time, and research effort to create, evaluate, and establish in everyday life, a critical question remains: which technological paths are the most promising? Furthermore, if this trend continues, it is necessary to understand how users will react to developments that are increasingly invasive.

Commercial examples of this shift include Meta's neural band using EMG, Samsung's Smart Ring for health data recording, and smart glasses from Ray-Ban which enable live translation and real-time image generation [27, 30]. However, these technologies do not have to stop at the body's surface. Complex technology is already prevalent in professional environments; for instance, exoskeletons are used to support humans in physically strenuous tasks or to improve tracking in virtual environments by transmitting accurate movement data directly from body parts [35, 3]. Beyond wearables, there is a path toward even closer integration via technologies

such as RFID and NFC chips, so-called "insideables", which are directly implanted under the skin to facilitate payment methods or smart features [31]. Disregarding the path of merely improving the human visually, these technologies aim to create permanent integrations with the human body, making interaction with technology a transient state that is more connected than ever before.

This trajectory reflects the transhumanist nature of technological development, specifically the goal of overcoming biological limitations [21]. The term itself specifies the question of human identity in the advent of technological developments, often aiming at the extension of human capabilities and the removal or overcoming of boundaries [17]. Bostrom describes this vision as creating the opportunity to live longer, healthier lives, enhance intellectual faculties, and achieve a greater degree of control over life [4]. Current developments echo this vision [1, 25]. However, given the rapid progress in this field, there is little research on how humans react to varying degrees of invasive technology [7] or the relationship between invasiveness and acceptance. Only when humans accept such technologies and integrate them into their self-image can the desired control become effective [4].

To investigate this relationship and provide an outlook on future developments, this research aims to analyze how humans perceive and accept these new technologies. By using an Extended Reality (XR) application, we immerse users in scenarios involving technology with different degrees of invasiveness, including stages that go beyond what is currently feasible. We propose three simulated technologies representing different levels of invasiveness to extract insights that can help direct the focus of future hardware development. Specifically, we seek feedback on the extent to which humans would be willing to adapt to possible future technological extensions.

The remainder of this paper is structured as follows: Section 2 provides an overview of related work. Section 3 details the design and methodology of the user study, including the planned three virtual device scenarios and the results of an explorative pre-study. Finally, Section 4 discusses these preliminary results and outlines the next steps for the main study.

2 RELATED WORK

Transhumanism represents a pluralistic spectrum of ideas, ranging from rational to visionary approaches. Influential movements include Extropianism, which advocates for continuous self-improvement through technology [5, 6], and the Hedonistic Imperative, which seeks the biological elimination of suffering. Singularitarianism, inspired by Ray Kurzweil [19], focuses on a potential intelligence explosion through artificial systems surpassing human intelligence [5]. The transhumanist discourse links technical strategies, from biotechnology and nanotechnology to mind uploading, with the goal of improving life and potentially achieving functional immortality [21]. A transhumanist perspective serves as a theoretical framework to contextualize current developments in body-proximate technologies. For instance, Popoveniuc et al. [26] locate Extended Reality (XR) within this framework, arguing that XR represents a transitional state between humanism and posthumanism by extending sensory and cognitive capabilities.

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Technologies such as XR and AR offer unique potential to experimentally investigate the influence of body-proximate technology and body modifications. This allows futuristic scenarios to be tested non-invasively during research and development. To capture the realism of virtual scenarios, the concept of *embodiment* is utilized. Embodiment occurs when properties of a virtual body are processed as if they were properties of the user's biological body [15]. Three dimensions contribute to this phenomenon: *Agency*, *Body Ownership*, and *Self-Location* [15]. AR allows users to perceive their real environment and themselves, supporting realism. Genay et al. [14] propose a "Body Avatarization Continuum" in AR, ranging from virtual accessories to full body avatarization. This helps locate the technologies used in this work on a conceptual scale. Furthermore, Nelson et al. [24] introduce the construct of "embodied empowerment," linking technology embodiment with a sense of empowerment, which serves as a significant predictor for adherence to body-proximate technologies.

While embodiment describes the extent to which technology is experienced as part of the body, technology acceptance addresses the willingness to actually use it. The Unified Theory of Acceptance and Use of Technology 2 (UTAUT2) extends previous models to voluntary consumer use [36]. For invasive, non-medical technologies, consumer-oriented determinants such as hedonic motivation (fun/experience) and social influence are crucial. De Andres-Sánchez et al. [8] and Arias-Oliva et al. [2] demonstrate that factors such as perceived usefulness, enjoyment, and social influence significantly impact the intention to use insides and wearables. While wearables are generally more accepted, prior studies indicate a hesitancy towards implants for health monitoring [16]. Initial approaches, such as those by Zhao et al. [42], explicitly connect embodiment with technology acceptance, showing that body awareness influences usage intention factors. This work systematically investigates this link between acceptance and embodiment relative to gradual invasiveness.

3 SCENARIOS AND PRELIMINARY STUDY

For the main study, three scenarios were designed to represent a tangible gradient of invasiveness. Specific functions were implied or mocked, all based on scientific literature, prototypes, and science fiction. The scenarios were implemented in an Extended Reality (XR) environment, with the goal of providing immersive interactions for the participants.



Figure 1: Excerpts from the scenarios presented in XR: (a) Cyber Arm Scenario, (b) Smart Bracelet, (c) Smart Contact Lenses

3.0.1 Smart Bracelet

This scenario features a *Smart Bracelet* that projects content directly onto the skin, distinguishing it from standard smartwatches. This concept is inspired by *iSkin* [37] and *Skin Buttons* [20], which utilize the skin as an input surface. Since wearables are socially established, this represents the lowest psychological entry threshold, positioning the skin as a subtle, accessible interface surface [34].

In the study implementation, the bracelet will be simulated using a physical fitness tracker on the participant's left arm to increase haptic realism, while content is projected onto the hand and forearm (see Figure 1b). The technical framework for the menu utilizes the

PalmMenu from the Meta XR Interaction SDK, enabling precise placement on the palm and intuitive interaction logic [23]. The interface includes elements designed to mimic realistic data:

- A heart rate monitor displays an average value between 70 and 80 bpm, varying randomly within this interval to suggest real-time measurement [10]. A step counter shows a stepcount and calorie burn chosen based on the average daily activity of young adults (18-34 years) [33].
- The display includes a weather tab with dummy data, a non-functional calculator inspired by iOS 18, and a calendar showing a fictitious schedule.
- To introduce a futuristic element, a floating hologram is anchored slightly above the palm. Users can manipulate its color, rotation, and geometric shape via a dedicated sub-menu. This feature aims to categorize the technology intuitively as advanced.

3.0.2 Smart Contact Lenses

The *Smart Contact Lenses* resemble standard lenses but with integrated technology where information is displayed directly in the user's field of view (see Figure 1c). This draws on real-world prototypes like those from XPANCEO [40] or Mojo Vision [11], and medical research on tear fluid monitoring [41]. This scenario represents medium invasiveness as it touches a sensitive sensory organ and continuously alters visual perception.

In the XR simulation, the impression of smart lenses is created through a "80s retro-futuristic" aesthetic (green lines and text) to match current prototype descriptions [32]. Two types of widgets are employed:

- Peripheral widgets are fixed to the edge of the field of view. These non-interactive elements display date, time, and biometric data carried over from the bracelet scenario. Additionally, a blood sugar value is displayed, modeled after values found in [12].
- Spatial widgets are placed within the environment and activated by gaze. Users dwell on an object to trigger a selection (visualized by a green loading circle).

Key features include a room analysis tool that labels objects (e.g., walls, tables, sofas) in real-time, and an electrolyte monitor displaying levels based on medical baselines [12]. A navigation feature projects arrows at half-body height towards an arbitrary target, simulating in world pathfinding. A gyroscope visualizes current head rotation, and a mocked camera function is implied via a "blink to capture" mechanic.

3.0.3 Cyber Arm

For the third scenario, we chose a robotic arm, totally replacing the biological left arm. Unlike medical prosthetics, this focuses on voluntary amputation for enhancement, a trope found in media and pop-culture like *Cyberpunk 2077* [39] or the Marvel Universe [13]. While commercial "elective" prosthetics are not yet available, research into "Cyborg products" suggests that demographic and psychological factors influence the willingness to undergo such invasive procedures [28, 29]. This scenario represents the highest invasiveness due to the permanent, structural alteration of the body.

In XR, a robotic arm is overlaid onto the participant's real arm using the Meta Movement SDK to map biological movement to the avatar via inverse kinematics [22] (see Figure 1a). Participants will wear a dark fabric sleeve to reduce the visual presence of their real arm and provide a haptic sense of "foreignness" or compression. The core interaction focuses on strength variation. Users will be

asked to interact with a virtual tablet to adjust the arm's power setting. To visualize the consequences of "superhuman" strength, the physics engine dictates how objects react:

- With low strength settings, users can lift light objects like a cup or glass safely.
- On higher strength settings, heavier objects (e.g., a bottle or a stone column) become liftable, but sensitive objects like a glass or cup will shatter upon interaction demonstrating excessive force.

To further enhance the illusion of improved strength, a physical prop manipulation is used: participants lift a heavy box before the simulation, which is surreptitiously swapped for a visually identical but much lighter box once the virtual arm is active.

4 PRE-STUDY

The primary objective of this research is to understand the interplay between the degree of technological invasiveness, the user's sense of embodiment, and the resulting technology acceptance. It is necessary to explore how different levels of physical integration, from external wearables to internal implants, affect the user's perception of the self and the device.

To ensure that the chosen scenarios elicit sufficiently distinct reactions regarding invasiveness and acceptance prior to a main study, we conducted an online pre-study. This pre-study serves to validate the presented scenario designs and gather preliminary data on user expectations and outlook on the technology choices. The general assumption guiding this preparation is that increased invasiveness requires a higher degree of embodiment to be accepted, yet may initially trigger lower acceptance rates due to perceived physical violation or social stigma.

4.1 Explorative Online Questioning

The pre-study was conducted as an online survey, introducing the three technology scenarios. For this survey, scenarios were presented in a shortened written form supplemented by images generated with GPT-5.0, to visualize the potential future devices in use. The survey included demographic questions, the Affinity for Technology Interaction (ATI/TAE-G-S) scale [38], and the UTAUT2 questionnaire [36] to assess technology acceptance.

4.1.1 Results

The survey was done to provide initial assessments of the scenarios' acceptance. We recruited participants online and via local lists, and evaluated ethical feasibility with our local ethics commission. A total of $N = 37$ participants completed the survey (17 female, 19 male, 1 diverse). The age distribution was skewed towards a younger demographic, with the majority ($n = 33$) aged between 18 and 34 years (18–24: $n = 23$; 25–34: $n = 10$). Older age groups were less represented (35–44: $n = 2$; 45–54: $n = 1$; 55+: $n = 1$). To understand the participants' general disposition towards technology, we assessed the TAE-G (*Technikaffinität zur elektronischen Gestaltung*) [18] questionnaire. The results indicated a relatively high affinity for technology within the sample ($M = 3.75$, $SD = 0.97$ (Highest Value is 5)).

Regarding prior experience with body-proximate technologies, almost all participants reported owning Bluetooth headphones ($n = 36$), and nearly half owned a smartwatch or fitness tracker ($n = 17$). However, ownership of emerging or invasive technologies was non-existent in this group; no participants reported owning smart glasses, smart rings, or implants ($n = 0$). Only one participant indicated owning none of the listed devices. This distribution confirms that while the participants possess a high affinity for technology, the specific scenarios involving smart visual aids and invasive modifications represent novel experiences for this cohort.

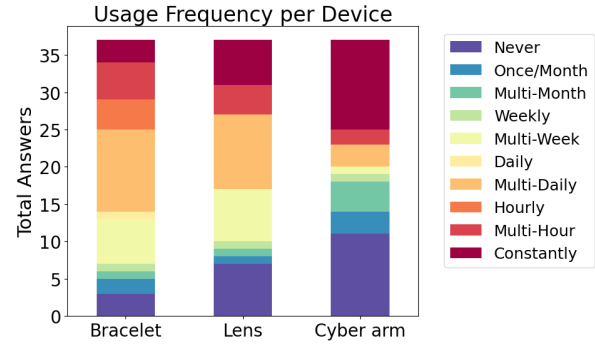


Figure 2: Participants' reported usage frequency when asked how often they would imagine using the technology.

Acceptance Rankings The UTAUT2 overall score was highest for the Smart Bracelet scenario ($M = 5.16$, $SD = 1.61$), followed by the Smart Contact Lens scenario ($M = 4.71$, $SD = 1.77$), and finally the Cyber Arm scenario with the lowest value ($M = 3.90$, $SD = 1.92$). Across all scenarios, the "Effort Expectancy" subscale achieved the highest values (indicating low perceived effort), while "Social Influence" received the lowest ratings.

Usage Frequency Distinct differences were observed in projected usage frequency in Figure 2. While over half of the respondents ($n = 23$) stated they would use the bracelet several times a day, the Contact Lenses showed a split distribution. The Cyber Arm showed the most polarized results, with significant groups indicating either constant use or total rejection.

These results indicate that the conceived scenarios generate sufficiently differentiated acceptance evaluations and are suitable for a follow-up main study.

5 DISCUSSION AND FUTURE WORK

The results of the preliminary survey confirm that the selected scenarios are perceived as distinct categories regarding their technological nature and potential impact on the user. This validation is crucial for the subsequent phase of our research, as it allows us to utilize these specific technologies to facilitate a tangible transition between different levels of invasiveness.

While the current market offers a plethora of body-proximate devices, and the scope of transhumanist discourse includes a vast array of potential enhancements, we restricted our design to these three specific technologies to ensure experimental feasibility. By locking the study to these distinct points on the invasiveness spectrum, we can effectively map the trajectory from socially established wearables such as the Bracelet to structural modifications like a Cyber Arm. This gradient aligns with central themes in transhumanism, exploring the shift from using technology as a tool to integrating it as a part of the biological self.

Building on these initial findings, in a future study, we aim to evaluate key metrics, specifically focusing on the dependency of *Acceptance* on the level of *Invasiveness*. Furthermore, we will investigate the role of *Embodiment*. A research question we aim to test is whether a higher degree of embodiment correlates with increased acceptance of invasive body-proximate technologies, i.e. if users are more likely to accept a highly invasive device if the simulation successfully tricks the brain into perceiving it as a natural part of the body.

The implementation for all scenarios is realized in Unity using the Meta Quest 3 headset. This device utilizes passthrough to facilitate XR. While these scenarios could theoretically be simulated using AR glasses, we deliberately chose a passthrough-VR headset.

The video passthrough can allow for a higher degree of immersion through the inherent nature of VR and robust visual occlusion. This is particularly relevant for the Cyber Arm scenario, where the user's real limb must be visually replaced onto the robotic counterpart. We anticipate that this setup will enable participants to better immerse themselves in the simulation, experiencing a more authentic use of the presented technologies.

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